

AnveshakHub Ecosystem

Complete System Specifications & Architecture Blueprint

This specification document details the user journeys, functional specifications, and technical architecture design for the AnveshakHub Ecosystem Platform. It outlines core interactions across multiple stakeholders - namely Industries, Experts, and Students - and provides the blueprint for the web administration panel, mobile client applications, relational data structures, and supporting services.

PREPARED BY:

Pranav SP & Vishnu SH

TARGET ENVIRONMENT:

Flutter Mobile | React Web Admin | NestJS Backend | PostgreSQL

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PART I: Stakeholder Analysis and User Journey

1. Purpose of the Document

The purpose of this document is to identify the stakeholders involved in the AnveshakHub ecosystem and define their journey within the proposed platform.

The platform involves multiple types of users, each having different responsibilities and permissions. Understanding their journey helps in designing a system that aligns with the actual workflow of AnveshakHub.

2. Stakeholders

The proposed platform consists of four primary stakeholders.

1. Industry
2. Expert
3. Student
4. AnveshakHub Administration

Each stakeholder interacts with the platform differently.

3. Industry User Journey

Step 1: Registration

The industry creates an account by providing:

- Company Name
- Company Type
- Industry Domain
- Company Website
- Contact Person Details
- Official Email Address
- Contact Number
- Company Registration Documents

After registration, a company profile is created.

Step 2: Profile Completion

The industry completes its profile by providing:

- Company Description
- Services Offered
- Current Challenges
- Technology Requirements
- Preferred Solution Areas

Step 3: Problem Statement Submission

The industry submits a problem statement.

Information may include:

- Problem Title

- Problem Description
- Expected Outcome
- Preferred Timeline
- Required Expertise
- Supporting Documents

Step 4: Approval Process

After submission, the problem statement enters review. At this stage:

- Industry can submit requirements.
- Industry cannot directly approach experts.
- Expert access remains restricted.

The AnveshakHub team reviews the request.

Step 5: Discussion and Meetings

The AnveshakHub team may:

- Contact the industry
- Schedule meetings
- Understand requirements
- Evaluate feasibility

Multiple meetings may occur before approval.

Step 6: Client Activation

If approved:

- Industry becomes an active client.
- Expert access is enabled.
- Project workflow becomes available.

If rejected:

- Request remains closed.
- Industry can submit future requests.

Step 7: Project Monitoring

Once a project begins, the industry can:

- View project progress
- View milestones
- Attend meetings
- Upload documents
- Review deliverables

4. Expert User Journey

Experts may include:

- Professors
- Technical Professionals
- Consultants
- Researchers
- Domain Experts

Step 1: Registration

The expert creates a profile. Information includes:

- Name
- Expertise
- Experience
- Qualifications
- Certifications
- Publications
- Previous Projects

Step 2: Verification

The AnveshakHub team verifies the expert profile before activation. Only approved experts become active users.

Step 3: Project Discovery

Experts can only view projects that have been approved by the AnveshakHub team. Unapproved requests remain hidden.

Step 4: Project Acceptance

Experts can:

- Review project details
- Evaluate requirements
- Accept participation

Step 5: Team Creation

After accepting a project, experts can:

- Create internship opportunities
- Define student requirements
- Define team size

Example: Maximum Students = 30

Applications remain open until the limit is reached.

Step 6: Student Evaluation

Experts can:

- Review applications
- Evaluate resumes
- Accept candidates
- Reject candidates

If students leave or are removed, available slots automatically reopen.

Step 7: Project Execution

Experts guide:

- Students
- Technical Teams
- Research Activities

while tracking project progress.

5. Student User Journey

Step 1: Registration

Students create profiles. Information includes:

- Academic Information
- Skills
- Resume
- Certifications
- Project Experience

Step 2: Profile Building

Students continuously improve their profiles by adding:

- Skills
- Certifications
- Achievements
- Completed Projects

Step 3: Opportunity Search

Students can view available opportunities created by experts. Examples:

- Internships
- Research Projects
- Industry Projects

Step 4: Application

Students apply to opportunities by submitting:

- Resume
- Supporting Documents
- Profile Information

Step 5: Selection

Experts review applications and make selections. Selected students join project teams.

Step 6: Participation

Students contribute to:

- Development Activities
- Research Tasks
- Documentation
- Project Deliverables

under expert supervision.

6. AnveshakHub Administration Journey

The administration team controls the overall platform.

Responsibilities

User Verification

Approve:

- Industries
- Experts
- Students (if required)

CRM Management

Track:

- New Industries
- Meetings
- Discussions
- Approval Status

Project Governance

Monitor:

- Active Projects
- Experts
- Teams
- Progress

Platform Administration

Manage:

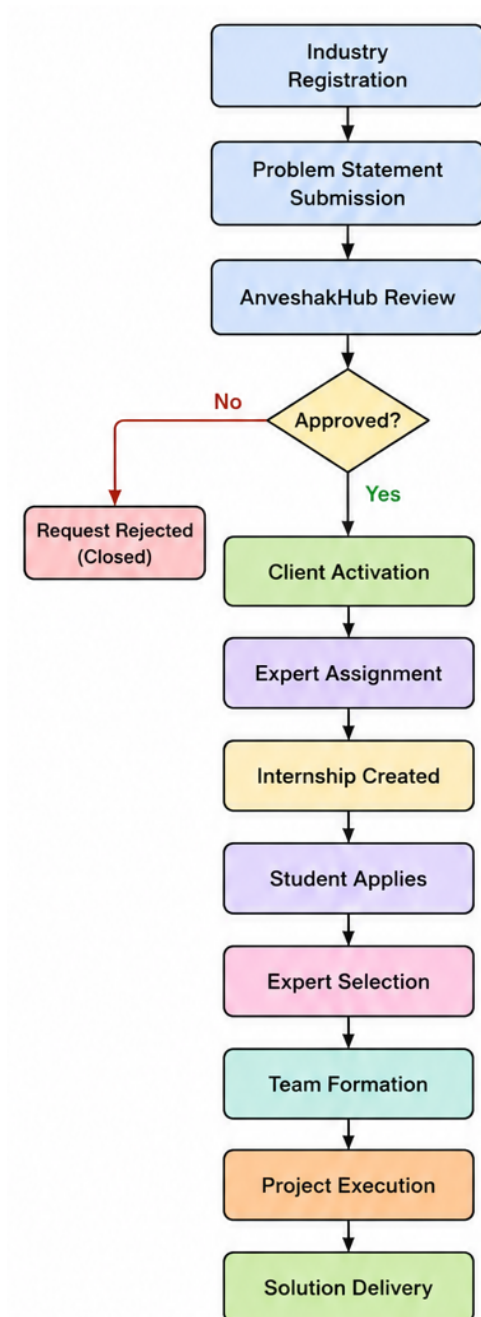
- User Accounts
- Permissions
- Reports
- Platform Operations

7. Summary

The platform is designed around four key stakeholders: industries, experts, students and the AnveshakHub administration team. Each stakeholder has a clearly defined journey and permission level. This structure helps ensure that project execution remains organized while allowing AnveshakHub to maintain control over approvals, expert engagement and overall project governance.

8. USER INTERACTION FLOW

The following flowchart represents the complete end-to-end user interaction and project lifecycle within the AnveshakHub Ecosystem.



This flow ensures a structured and transparent process from industry registration to final solution delivery through expert and student collaboration, governed by AnveshakHub.

PART II: Functional Requirements Specification (FRS)

1. System Components

The AnveshakHub Ecosystem consists of two primary system interfaces:

1.1 Mobile Application

Purpose:

Provide a centralized platform for Industries, Experts, and Students.

Key Features:

- User Registration & Authentication
- Profile Management
- Project Participation
- Internship Management
- Notifications & Updates
- Document Upload & Access

1.2 Web Administration Panel

Purpose:

Provide complete administrative control over platform operations.

Key Features:

- User Verification
- CRM Management
- Project Governance
- Meeting Management
- Analytics & Reporting
- Platform Monitoring

2. Authentication Module

Purpose:

Secure access management for all stakeholders.

Functional Requirements:

- User Registration
 - Industry Registration
 - Expert Registration
 - Student Registration
- Login Management
 - Email Authentication
 - Password Encryption
 - Session Management
- Password Recovery
 - Forgot Password
 - Reset Password

- OTP/Email Verification
- Account Verification
 - Approve Users
 - Reject Users
 - Suspend Accounts
 - Reactivate Accounts

3. Industry Module

Purpose:

Enable industries to submit challenges and monitor project execution.

Functional Requirements:

A. Industry Dashboard

- Company Information
- Active Projects
- Project Status
- Notifications

B. Problem Statement Management

- Create Problem Statements
- Edit Problem Statements
- Upload Supporting Documents
- Track Approval Status

C. Meeting Management

- Request Meetings
- View Meeting Schedule
- Access Meeting Notes

D. Project Monitoring

- View Milestones
- Monitor Team Progress
- Review Deliverables
- Download Reports

4. Expert Module

Purpose:

Enable experts to guide projects and manage student teams.

Functional Requirements:

A. Profile Management

- Expertise Details
- Qualifications
- Certifications
- Previous Projects

B. Project Participation

- Browse Approved Projects
- Accept Assignments

- View Requirements
- Track Progress

C. Opportunity Management

- Create Internships
- Edit Opportunities
- Close Opportunities
- Manage Applications

D. Student Evaluation

- Resume Review
- Candidate Selection
- Candidate Rejection
- Team Formation

E. Project Monitoring

- Assign Tasks
- Update Milestones
- Submit Reports
- Track Performance

5. Student Module

Purpose:

Enable students to participate in industry projects and internships.

Functional Requirements:

A. Profile Management

- Resume Upload
- Skills Management
- Certifications
- Achievements

B. Opportunity Discovery

- Browse Internships
- Browse Research Projects
- Filter Opportunities
- View Deadlines

C. Application Management

- Submit Applications
- Upload Supporting Documents
- Track Application Status

D. Project Participation

- View Assigned Tasks
- Submit Deliverables
- Track Progress
- Communicate with Experts

PART III: System & Technology Architecture

1. Architecture Overview

The proposed platform utilizes a centralized, secure architecture where mobile applications and the web administrator panel communicate with NestJS backend services. Relational data models are managed using PostgreSQL, while file assets and notifications are routed through independent external services.

2. Tech Stack Selection Summary

Component	Technology Chosen
Mobile Application	Flutter (Single codebase for Android & iOS)
Web Administration Panel	React.js with TypeScript
Backend Services	NestJS (TypeScript)
Database Layer	PostgreSQL with Prisma ORM
Authentication	JWT Authentication + Refresh Tokens + RBAC
File Storage Layer	Cloudinary -> Migrating to AWS S3
Push Notifications	Firebase Cloud Messaging (FCM)
Email Service Provider	Resend or SendGrid
Future AI Services Layer	Python FastAPI

3. Future Architecture Enhancements

- AI recommendation modules to suggest matching experts for problem statements.
- Automated transcription and summaries for scheduled CRM discussion meetings.
- Intelligent project scoring metrics to track team execution performance.
- Integration endpoints to external enterprise software suites.

This complete specification serves as the conceptual blueprint for database schema design, REST API development, UI/UX mockups, and mobile client implementation.

--- End of Specifications ---